



ST JOSEPH ENGINEERING COLLEGE

An Autonomous Institution
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INTERNET ENABLED PROTOTYPING (22IEP38)

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Course Project - Smart Library

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Problem Statement:

In traditional libraries, users often find it difficult and time-consuming to locate books after searching them in a catalog, especially in large libraries with multiple shelves and similar classifications. This leads to confusion, overcrowding, and frequent dependence on library staff for assistance. To overcome this problem, a Smart Library system is proposed in which, when a user selects a book from the displayed list, a light blinks on the shelf where the book is kept, enabling quick and accurate book location while improving user convenience and library efficiency.

SDGs linked:

- SDG 4: Quality Education
- SDG 9: Industry, Innovation, and Infrastructure
- SDG 8: Decent Work and Economic Growth

Introduction:

This project comes under the domain of IoT-based automation in library management systems. Traditional library systems, although equipped with digital catalogues, still require users to manually search for books on physical shelves. This process is time-consuming and often leads to confusion, especially in large libraries. Currently available solutions such as RFID-based library systems provide accurate tracking and management of books, but they are expensive and involve complex infrastructure, making them unsuitable for small- and medium-scale libraries. Therefore, there is a growing need for a simple, low-cost, and efficient smart library system that utilizes IoT technology to assist users in quickly locating books. By using automation and visual indicators, the proposed system aims to reduce search time, improve user experience, and enhance overall library efficiency.

Objectives

- To design and develop an IoT-based smart library system for efficient book location.
- To reduce the time and effort required by users to find books in library shelves.
- To provide a low-cost alternative to complex RFID-based library management systems.
- To implement automation with visual indicators for easy identification of book locations.
- To improve user experience and operational efficiency in libraries.
- To ensure the system is simple, scalable, and easy to maintain.

Methodology

- Step 1: Requirement Analysis**
 - Hardware: ESP32, LEDs
 - Software: Arduino IDE, Web or Mobile App

- Step 2: System Design**

- The system was configured as a **Wi-Fi Access Point** (SSID: "Smart Library"), creating a local network that allows users to access the interface via the IP address **192.168.4.1** without needing the internet.

- Step 3: Working Process**

- A database of 20 books was created, where 6 available books are mapped to specific physical Rows and GPIO pins (13, 12, 14, 27, 26, 25).

- Step 4: Implementation**

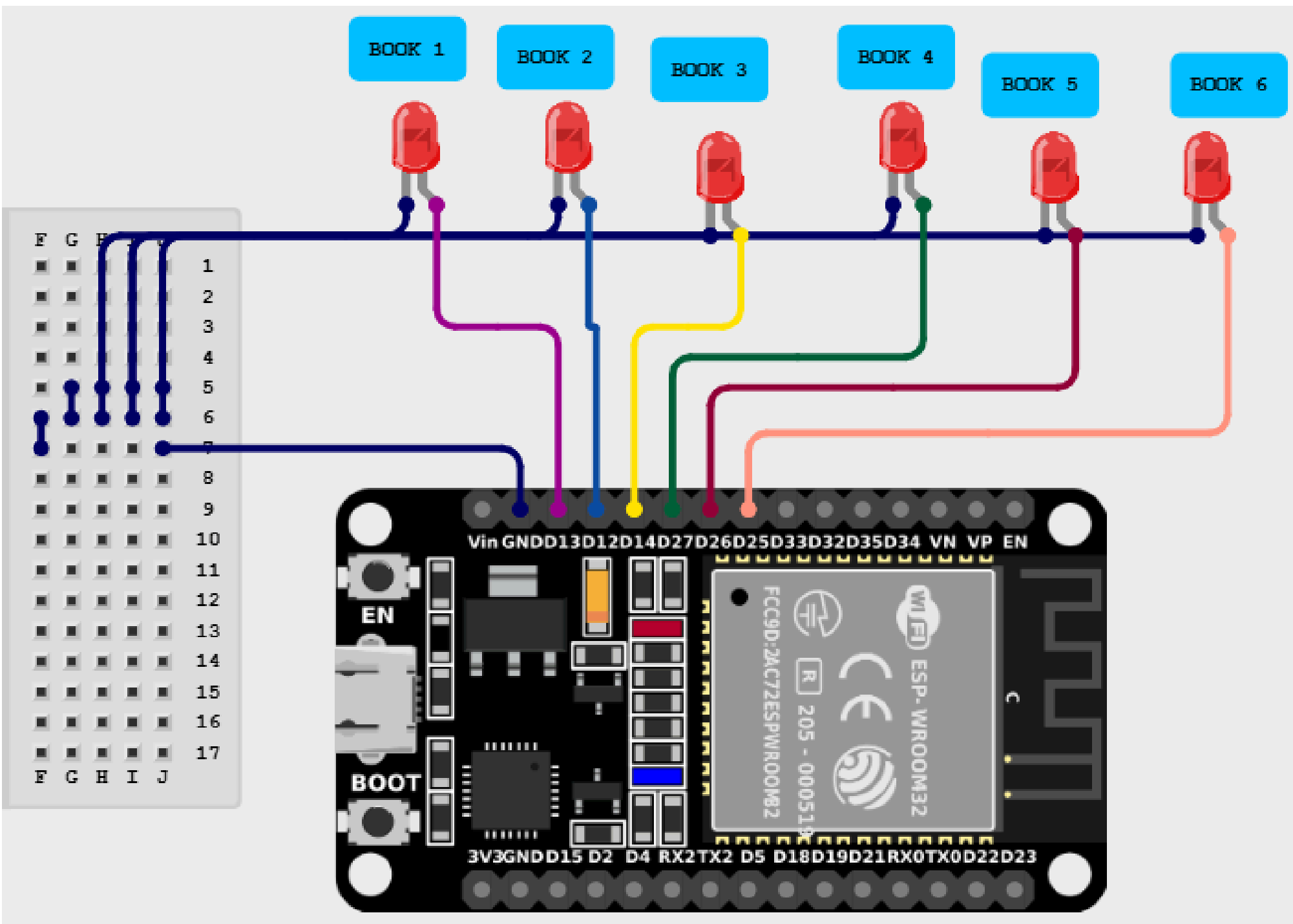
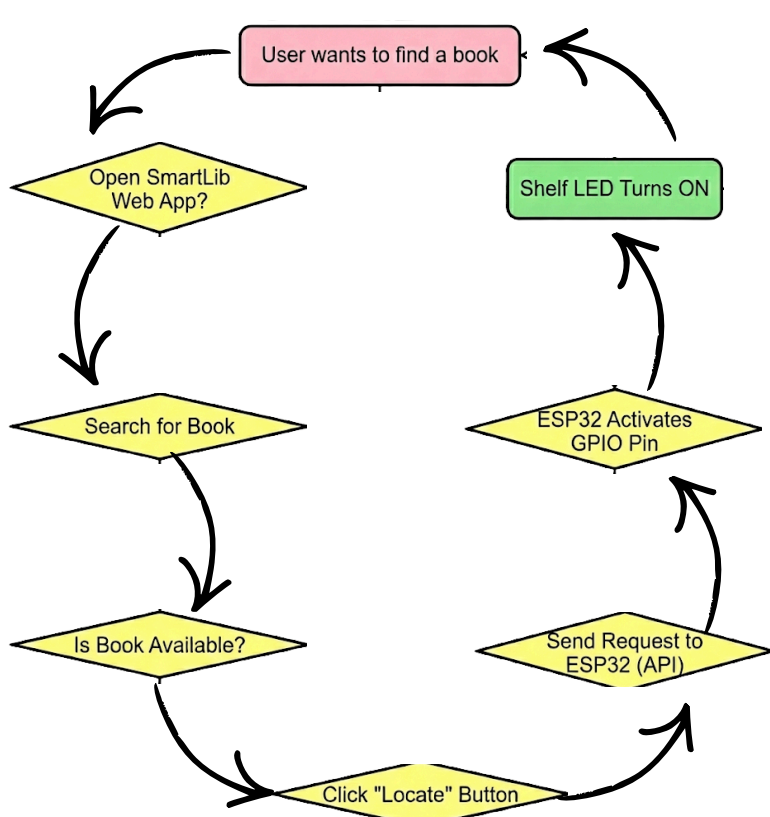
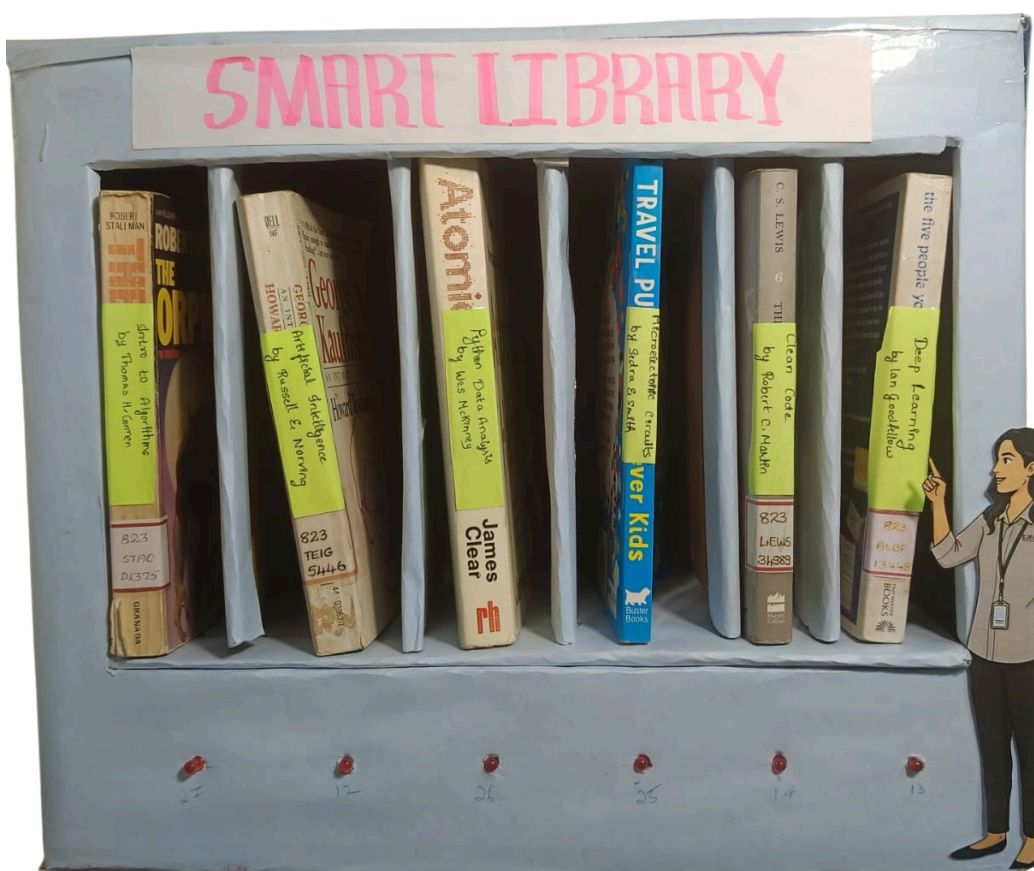
- Connect ESP32.
- Code the microcontroller.
- Create and link database.
- Build app or web interface.
- Integrate and test the system.

- Firmware was developed to host the "Offline" web interface and manage logic such as the **10-second LED timeout** and a **2-minute idle reset** that returns the app to the Home Screen.

- Step 5: Testing**

- The prototype was verified by ensuring that **selecting a book on the dashboard correctly triggered the corresponding physical LED** on the shelf and displayed the "More Details" popup accurately.

Model/sketch/circuit diagram



Results and discussion and images

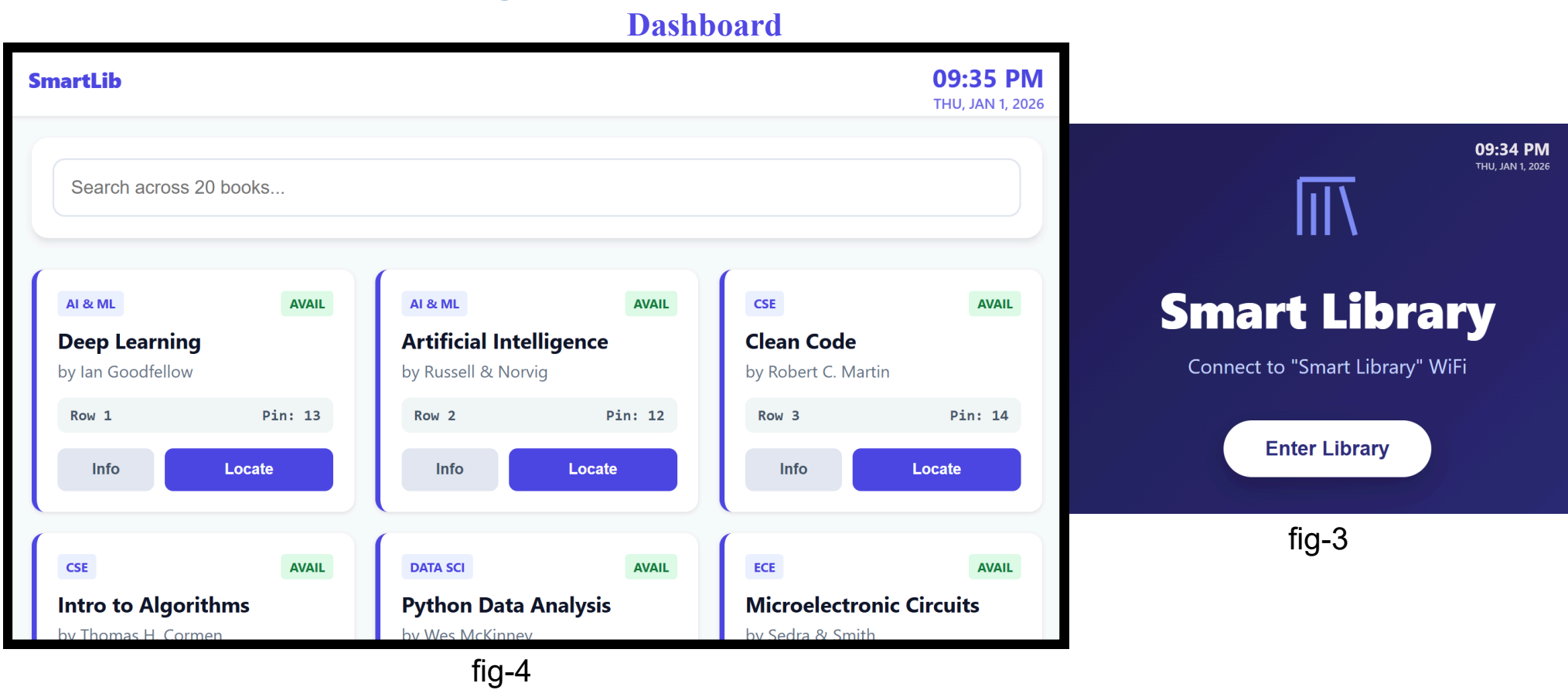


Fig-3 (Home Screen): The user's entry point. A clean, branded landing page that welcomes the user and provides a simple "Enter Library" button to access the dashboard.

Fig-4: The SmartLib Dashboard The Dashboard is the central control interface, providing users with a **searchable database** of books that displays titles, authors, and real-time availability status. Users can **verify if a book is in stock** and click the "Locate" button to trigger a specific LED on the physical shelf. This layer effectively bridges the digital software with the physical hardware, **translating user clicks into real-world light signals**.

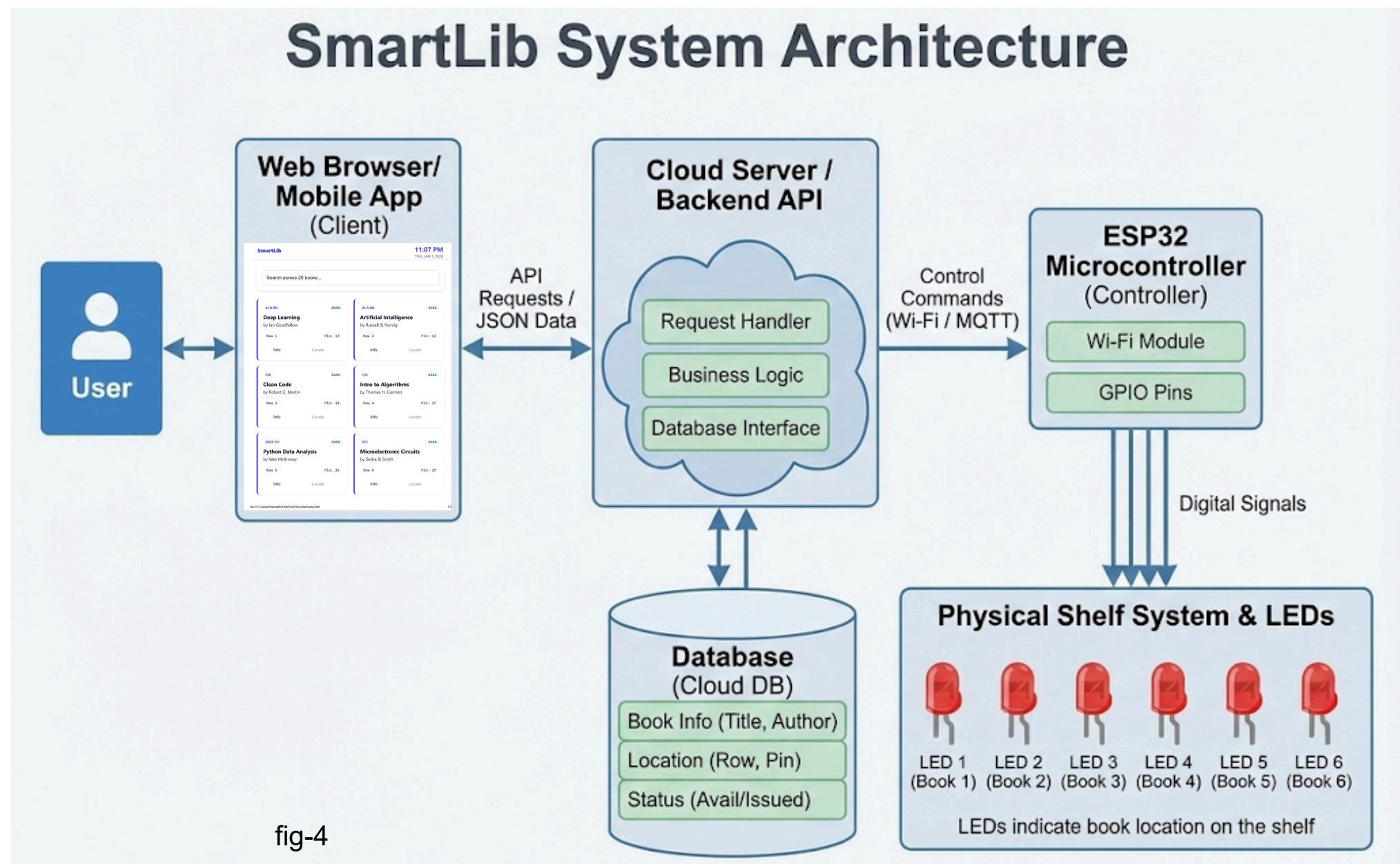


Fig-5: **System Architecture Diagram** This diagram provides a high-level technical overview of how the hardware and software components communicate within the **Smart Library system**. It visualizes the data flow starting from the client-side web browser, which sends **HTTP requests** to the **web server hosted directly on the ESP32 microcontroller**. The ESP32 processes these requests using its **internal backend logic to determine which GPIO pin** corresponds to the selected book. Finally, the diagram shows the output stage, where **digital signals trigger specific LEDs** on the physical shelf to guide the user to their desired book.

Conclusion

The implementation of a smart library system incorporating visual cues, such as blinking LEDs on shelves, successfully addresses the traditional challenge of finding physical books in a large collection. By leveraging modern technologies like the Internet of Things (IoT) this system provides real-time book positioning and a user-friendly interface (e.g., mobile app or web application) that guides users directly to the correct shelf. This automation not only enhances the overall user experience and satisfaction but also streamlines library operations, saves significant time for staff and users, and allows library professionals to focus on more complex tasks. The integration of physical library materials with digital management ensures a transparent and highly efficient system for book tracking and inventory control, ultimately modernizing library services and making them more accessible and efficient.

Reference

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